CN SDNmininet project (Orange)

Problem Statement :
Automatically detect and maintain a list of hosts in the SDN network.

Execution Steps:

1.Start the POX controller using

```
./pox.py log.level --DEBUG host_discovery
```

2. Open a new terminal and create network topology using Mininet

```
sudo mn --topo single,3 --controller remote
```

3. Test connectivity between hosts:

```
Pingall
```

4. Generate traffic to trigger host discovery:

```
h1 ping -c 3 h2
```

5. Observe the controller terminal for output showing discovered hosts.

OUTPUT:

```
greeshma@greeshma-VirtualBox:~/cn_orange$ sudo mn --topo single,3 --controller remote
[sudo] password for greeshma:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.48 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.24 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=1.33 ms

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.236/1.681/2.484/0.568 ms
mininet>
```

```
greeshma@greeshma-VirtualBox:~/cn_orange/pox$ cd ..
greeshma@greeshma-VirtualBox:~/cn_orange$ rm -rf pox
greeshma@greeshma-VirtualBox:~/cn_orange$ git clone https://github.com/noxrepo/pox.git
Cloning into 'pox'...
remote: Enumerating objects: 13425, done.
remote: Counting objects: 100% (277/277), done.
remote: Compressing objects: 100% (50/50), done.
remote: Total 13425 (delta 258), reused 227 (delta 227), pack-reused 13148 (from 2)
Receiving objects: 100% (13425/13425), 5.14 MiB | 1.64 MiB/s, done.
Resolving deltas: 100% (8704/8704), done.
greeshma@greeshma-VirtualBox:~/cn_orange$ cd pox
greeshma@greeshma-VirtualBox:~/cn_orange/pox$ ls
debug-pox.py  ext  LICENSE  NOTICE  pox  pox.py  README.md  setup.cfg  tests  tools
greeshma@greeshma-VirtualBox:~/cn_orange/pox$ cd pox
greeshma@greeshma-VirtualBox:~/cn_orange/pox/pox$ nano host_discovery.py
greeshma@greeshma-VirtualBox:~/cn_orange/pox/pox$ cd ..
greeshma@greeshma-VirtualBox:~/cn_orange/pox$ python3 pox.py log.level --DEBUG host_disco
very
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:host_discovery:🚀 Host Discovery Service Started Successfully
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.12.3/Mar 3 2026 12:15:18)
DEBUG:core:Platform is Linux-6.14.0-37-generic-x86_64-with-glibc2.39
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
INFO:host_discovery:Switch 1 connected
INFO:host_discovery:✅ New host discovered: 3e:0f:48:87:4c:35 on port 3
INFO:host_discovery:✅ New host discovered: 6a:c8:ab:ff:37:92 on port 1
INFO:host_discovery:✅ New host discovered: 5a:33:1a:d5:c3:9f on port 2
DEBUG:openflow.of_01:1 connection aborted
```